Unit D1 Decision Mathematics 1

IAS optional unit for Mathematics and Further Mathematics Externally assessed

D1.1 Unit description

Algorithms; algorithms on graphs; the route inspection problem; critical path analysis; linear programming; matchings.

D1.2 Assessment information

Preamble

Students should be familiar with the terms defined in the glossary attached to this specification. Students should show clearly how an algorithm has been applied. Matrix representation will be required but matrix manipulation is not required. Students will be required to model and interpret situations, including cross-checking between models and reality.

Examination

The examination will consist of one $1\frac{1}{2}$ hour paper. It will contain about seven questions of varying length. The mark allocations per question will be stated on the paper. All questions should be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π, x^2, $\sqrt{x}$, $\frac{1}{x}$, x^y, $\ln x$, e^x, $x!$, and memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.